
Summary

This document is a guide for National Healthcare Safety Network Collaborative ([NHSNCoLab](#)) Cerner facilities to create Cerner extracts to pull populations of patients and output a pipe delimited file. The extracts can be installed by NHSNCoLab's Cerner client base to reduce the effort on health systems and speed up the implementation. The extracts are designed to pull all in-house patients, and they can be set-up to run on a schedule in Olympus. There may be some site-specific changes needed based on the Cerner client's specific set-up or workflow.

Cerner Prerequisites

The extracts can be used for any Cerner facility that is running Cerner Millennium. The only exception is for facilities that are in a multi-tenant environment. The multi-tenant set-up involves multiple disparate organizations that share one physical Cerner Millennium domain with logic data separation. Because these environments house data from multiple disparate organizations the access to data is tightly controlled and only DA2 reporting is available. Cerner's multi-tenant facilities tend to be smaller hospitals that are looking to reduce costs by sharing a Cerner Millennium environment. Most health systems will have their own Cerner environment.

These extracts can be run regardless of who hosts the Cerner Millennium environment. Cerner clients can either be client hosted (on-prem) or Cerner hosted. In either case the installation and execution of the extracts are the same. It's more common to run into facilities that are Cerner hosted as facilities have been transitioning away from hosting their own environments.

Cerner Client Resource Requirements

Skill Set	Install step	Estimated time
Cerner CCL Resource	- Review code and save the base program in the appropriate environment - Update with site specific configuration - Start pulling test extracts as part of the testing process - Move program to prod and execute program as needed	5-10 hours
sFTP Resource	Set-up sFTP job to send files to NHSN on a scheduled basis. This typically uses an established sFTP process that is already set-up at the health system.	2-4 hours
Olympus Resource	Set-up ops job schedule in test and production for the extracts.	2-4 hours
Client Analyst	Assist with testing and validation process to ensure populations are getting pulled for	5-10 hours

	the correct facilities, nurse units, etc. This resource would review the extracts and compare with patients in PROD/TEST.	
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Install Instructions

The following items need to be set up within Cerner for the extracts to run.

1. CCL Program Installation
 - a. Open Discern Visual Developer (DVD) and save the '1_jh_lantana_inhouse_ext.prg' source code files in the appropriate domain(s). The source code should be saved in the directory the facility has designated for custom CCL programs (e.g. cust script). This can vary by facility so it's best to check with the facility where they want this to be stored.
 - b. Review the script and make any site-specific changes that are needed.
 - c. Compile the program in Discern Visual Developer.

Program Execution

The following is an explanation of the prompt variables that are set-up to be passed into the program to control what information is extracted and how it's extracted.

1_jh_lantana_inhouse_ext.prg:

PROMPT

"Output Directory:" = "CCLUSERDIR:",

"Debug Flag (Y/N)" = "N",

"Facility Display" = "Facility Display Value"

WITH OUTDEV, DEBUG_FLAG, FAC_DISP

Output Directory: Specify the directory the extract files should be written to. This can be any directory that is accessible from the Cerner Millennium environment.

All the files will be output using the following naming convention and time stamped with the current date and time:

lantana_census_extract_YYYYMMDD_HHMMSS.dat

Debug Flag (Y/N): If the debug flag is set to 'Y' then call echo statements will be logged out in Discern Visual Developer and the file will not be written out. This is helpful for testing and troubleshooting. If the debug flag is set to 'N' then the call echo statements will be suppressed, and the file will be written out to the directory specified.

Facility Display: This is the display value of the facility on code set 220 that needs to be included in the extract.

Discern Visual Developer or Operations Job Execution

Below is an example execution that will pull all inhouse patients and output to a directory:

```
execute 1_jh_lantana_inhouse_ext:group1 "CCLUSERDIR:", "N", " Hospital Name Here" go
```

Site Specific Configurations

The following are site specific items that need to be adjusted based on site specific requirements:

- Patient type: the patient types being pulled in the query should be reviewed to make sure appropriate patient types are included in the file. This is fairly standard but can vary across facilities. Ensure that the patient types include all individuals classified as **Inpatient**, **Observation**, and **Emergency** patients.

Facilities: the facilities included in the query should be reviewed and adjusted for the specific facilities in scope. (In scope refers to only the facility/facilities associated with the reporting NHSN OrgID).

END